PLC Counter and Program Control Instructions

Pages 8–14 (Lecture Notes)

1. Counter Addressing in SLC 500

In Allen-Bradley SLC 500 PLCs, all counters are stored in a dedicated counter file.

- Counter file number: **File 5 (C5)**
- Counter address range: **C5:0 to C5:255**
- Each counter element shares a common base address

Counter Address Format

C5:3

- **C5** → Counter file number
- **:3** → Counter element number

2. Counter Status Bits

Counter status bits are stored in the control word and are used in ladder logic for decision-making.

Bit	Function
DN	Done bit (accumulated = preset)
CU	Count-up enable bit
CD	Count-down enable bit
OV	Overflow bit
UN	Underflow bit
UA	Update accumulator bit

The **DN (Done)** bit is the most frequently used counter status bit.

3. Count-Up Counter Configuration

A count-up counter (CTU) requires three values to be configured by the programmer.

Required Parameters

- **Counter Number** (e.g., C5:0)
- **Preset Value**
- **Accumulated Value**

Value Range

Preset and accumulated values are stored as 16-bit signed integers.

$$-32{,}768 \leq \text{Value} \leq +32{,}767$$

Reset Operation

- Reset instruction clears the accumulated value to zero
- DN bit becomes FALSE after reset

4. Motor Control Using an Up-Counter

Counters are often used to limit the number of machine operations.

Program Objective

- Allow motor operation only **10 times**
- Stop motor automatically after reaching preset count

Operation

- Each motor OFF-to-ON transition increments the counter
- Preset value = 10
- DN bit disables motor output using Examine-OFF instruction

The motor is controlled by the counter **DONE bit**, not directly by the counter instruction.

5. Reset and Reuse of Counter-Based Motor Control

- Reset pushbutton clears accumulated count
- DN bit becomes FALSE
- Motor operation is allowed again

Without reset, the motor remains disabled once the preset count is reached.

6. Can-Counting System Using Multiple Counters

Real industrial systems often require multiple counters working together.

Counters Used

- **C5:2** → Counts individual cans
- **C5:1** → Detects completion of one package (10 cans)
- **C5:3** → Counts total packages per day

Operation Flow

1. Sensor detects each can
2. Can counter increments
3. After 10 cans, package counter DN bit becomes TRUE
4. Internal bit initiates box-closing operation
5. Total package counter increments

The DN bit of one counter can control the operation of other counters.

7. Daily Reset Operation

- Pushbutton resets all counters
- Daily production count starts from zero
- Used for production tracking and reporting

8. Program Control Instructions

Program control instructions control the execution flow of a PLC program rather than field outputs.

Common Program Control Instructions

Instruction	Function
JMP	Jump to label
LBL	Label definition
JSR	Jump to subroutine
RET	Return from subroutine
SBR	Subroutine block
TND	Temporary end of program scan
MCR	Master Control Reset
SUS	Suspend execution

Program control instructions affect **logic execution**, not physical outputs.

9. Key Exam Points Summary

- Counters are stored in file C5
- DN bit indicates completion of counting
- Preset and accumulated values are 16-bit signed
- Counters are widely used for machine control
- Program control instructions manage execution flow